SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1025	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Problem Statement:	**Bus Ticket Reservation System**		
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Date:	10-10-2026	**Duration:**	60 minutes

Code of Conduct

I K Sampath Reddy (192424285) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *__K Sampath Reddy____*

Annexure A – Agile Sprint Planning & User Story Workshop

Bus Buddy: A Smart Bus Ticket Reservation System

Agile Sprint Planning and User Story Workshop: Product Vision, Backlog and Sprint 1 Plan

No.	Section (as required in Annexure A)	Key content
1	Identify the Project Vision and Stakeholders	Vision, objectives, stakeholder map
2	Create Epics and User Stories	8 epics, 32 user stories
3	Define Acceptance Criteria	Given–When–Then for Sprint 1 stories
4	Prioritize the Product Backlog	Value, MoSCoW, dependencies, ranking
5	Plan the Sprint Backlog	Sprint 1 stories, tasks, schedule
6	Estimate Effort Using Story Points	Planning Poker, Fibonacci, justification
7	Present the Sprint Goal and Expected Deliverables	Sprint goal, deliverables, review plan

1. Identify the Project Vision and Stakeholders

1.1 Product Vision

BusBuddy is a bus ticket reservation system for passengers and bus operators. The vision below follows the standard Agile vision-statement template, so every later decision in this workshop (stories, priorities and sprint goal) can be checked against it.

Vision Statement

For bus passengers and bus operators **who** lose time and money because of slow, manual ticket counters and double-booked seats,

Bus Buddy is a smart bus ticket reservation system **that** lets a passenger find a bus, choose a real seat, pay and receive a QR e-ticket in minutes.

Unlike manual counters and basic booking sites, **our product** locks every seat during payment so it can never be sold twice, protects women passengers with safe-seat rules, and gives operators live sales and occupancy data.

Vision in one line: every bus journey can be booked in three minutes, and no seat is ever sold twice.

1.2 Project Objectives

#	Objective	Measure of success (target)
1	Fast, simple booking	Median time from search to confirmed ticket is 3 minutes or less.
2	Zero double booking	No seat is sold twice during testing or after launch.
3	Reliable payments and refunds	At least 98% of valid payments succeed; the refund amount is shown instantly on cancellation.
4	Less manual work for operators	Counter paperwork reduced by about 60% within 6 months of launch.
5	Strong online adoption	At least 70% of seats sold online within 6 months of launch.
6	Fast and available system	Search results in under 3 seconds; 99.5% uptime.

1.3 Key Stakeholders

A stakeholder is anyone who uses, builds, funds, regulates or is affected by BusBuddy. Nine stakeholders were identified; the first six shape the product directly, and the rest influence it from outside.

Stakeholder	Role in the project	Main expectation	How we engage them
Passengers	End users who search, book, pay, travel and cancel.	Fast booking, real seat availability, safety, fair refunds.	Beta testing and feedback at Sprint Reviews.
Bus Operator / Admin	Customer and funder; manages buses, routes, fares and reports.	Control over schedules and fares, more sold seats, live reports.	Attends every Sprint Review; supplies the Product Owner.
Product Owner	Owns and orders the product backlog; accepts or rejects finished stories.	Maximum business value in every sprint.	Daily availability; backlog refinement each sprint.

Stakeholder	Role in the project	Main expectation	How we engage them
Scrum Master	Coaches the team in Scrum and removes obstacles.	A focused team and a protected sprint.	Facilitates all Scrum events.
Development Team	5 members (2 backend, 2 frontend, 1 QA) who design, build and test.	Clear stories, realistic commitments, stable scope.	Commits to the sprint goal; daily stand-up.
Conductors / Drivers	Scan QR tickets and update trip status.	A quick scanner that works on weak networks.	Feedback sessions before Sprint 5.
Payment Gateway Provider	External service that processes payments and refunds.	Correct, secure integration and compliance rules.	Follow integration guides; test in sandbox.
Transport Authority	Regulates fares, safety and records.	Accurate, tamper-proof trip and passenger data.	Share reports and compliance notes.
Customer Support Team	Handles passenger complaints and queries.	Admin tools to find bookings and fix issues quickly.	Review support needs before launch.

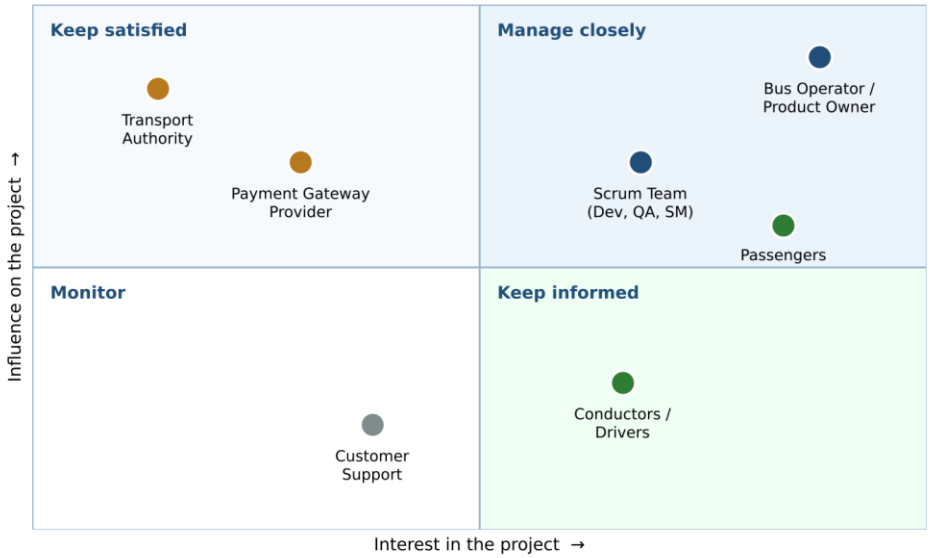


Figure 1: *Stakeholder map (influence versus interest)*

The map shows where effort should go. The Bus Operator / Product Owner, the Scrum Team and Passengers are managed closely, because they shape the product every sprint. The Transport Authority and the Payment Gateway Provider have high influence but low day-to-day interest, so they are kept satisfied through compliance and correct integration.

1.4 Assumptions and Constraints

- The project follows Agile Scrum with **2-week sprints** and six sprints (a 12-week release), as chosen in the earlier case study.
- The development team has five members and no earlier velocity data, so Sprint 1 estimates are based on capacity and Planning Poker.
- Payment, SMS and map services are bought from external providers and tested in sandbox mode first.
- Only domestic routes and one currency are covered in this release (see the project scope in the earlier case study).

2. Create Epics and User Stories

2.1 From Vision to User Stories

The vision was first split into **epics** (large pieces of value that take more than one sprint) and each epic was then broken into small **user stories** that fit inside a sprint. Every story uses the standard Agile format:

User story format

As a <user>, **I want** <feature>, **so that** <benefit>.

Example (US-01): As a passenger (new user), I want to register with my name, email, mobile number and password, so that I can create a personal account and book tickets securely.

Each story was checked against the **INVEST** rules: Independent, Negotiable, Valuable, Estimable, Small and Testable. Stories that were too big for one sprint were split.

2.2 Epics

Epic	Name	What it covers	Business value
E1	**User Accounts & Access**	Registration, secure login, password recovery and profile management.	Trust and secure access for every user.
E2	**Fleet, Route & Schedule Management**	Admin tools to set up buses, routes, trips and fares.	Gives the operator control; no trips means nothing to sell.
E3	**Search & Seat Selection**	Finding buses, viewing the live seat map and locking seats safely.	Core of the product; removes double booking.
E4	**Booking & Payment**	Traveller details, online payment, retry and e-ticket with QR.	Direct revenue and paperless travel.
E5	**Cancellation, Refund & Waitlist**	Online cancellation, automatic refunds and waitlist promotion.	Fairness, trust and fewer empty seats.
E6	**Notifications & Tracking**	SMS / email alerts, live bus tracking and delay alerts.	Fewer missed buses and fewer support calls.
E7	**Operations & Reporting**	Conductor QR scanning, trip status and admin dashboards.	Faster boarding and data-driven decisions.
E8	**Security, Quality & Operations**	Audit logs, hardening, load testing and monitoring.	Safe, fast and reliable at launch.

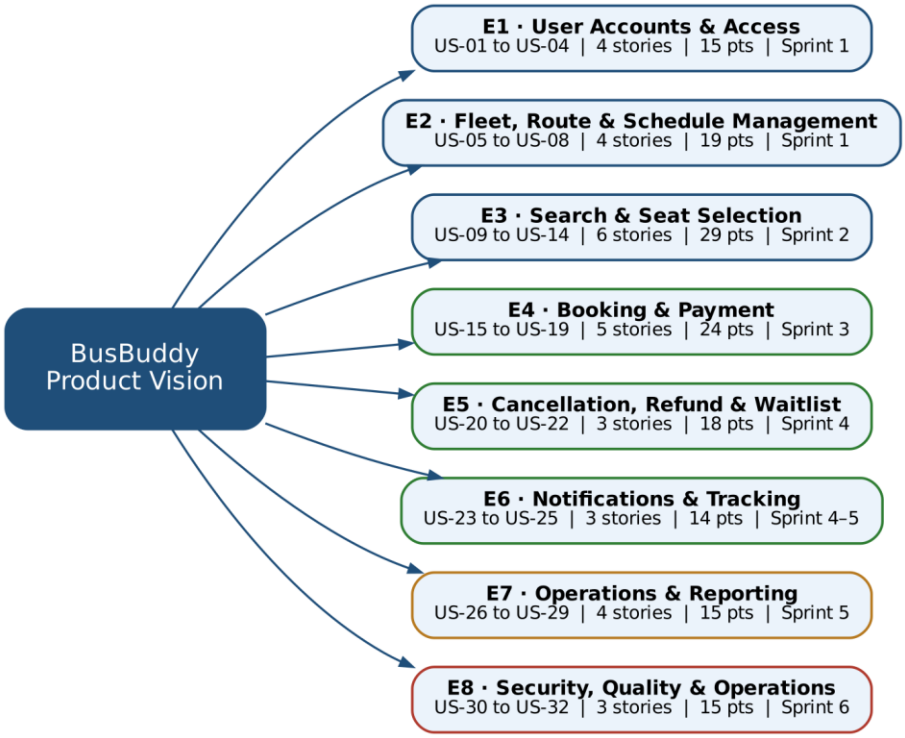


Figure 2: *Epic map: eight epics, with story count, points and target sprint*

2.3 User Stories by Epic

In total, 32 user stories were written (149 story points). They are listed below by epic, in the format As a <user>, I want <feature>, so that <benefit>.

E1 · User Accounts & Access			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-01	passenger (new user)	to register with my name, email, mobile number and password	I can create a personal account and book tickets securely
US-02	registered passenger	to log in using my password or a one-time password (OTP)	I can access my account quickly and safely
US-03	passenger	to reset a forgotten password using an OTP	I can regain access without contacting support
US-04	passenger	to view and update my profile details	my tickets and alerts always use correct contact information

E2 · Fleet, Route & Schedule Management			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-05	bus operator (admin)	role-based access to the admin panel	only authorised staff can change buses, fares and schedules
US-06	admin	to add, edit and deactivate buses with their seat layout	passengers see accurate bus types and seat maps
US-07	admin	to create routes with source, destination and boarding points	passengers can search and board at the right stops
US-08	admin	to schedule trips and set fares, including daily recurring trips	buses are available for booking without repeated manual entry

E3 · Search & Seat Selection			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-09	passenger	to search buses by source, destination and travel date	I can quickly find a suitable trip
US-10	passenger	to filter and sort results by price, departure time and bus type	I can compare options and choose the best one
US-11	passenger	to see a live seat map showing available, booked and ladies seats	I can pick the seat I prefer with confidence
US-12	passenger	the seats I select to be locked for 8 minutes	nobody else can book them while I complete payment
US-13	woman passenger	the system to avoid placing an unrelated male passenger beside my seat	I feel safe while travelling
US-14	passenger	to choose my boarding point from the route's stops	I can board at the stop nearest to me

E4 · Booking & Payment			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-15	passenger	to enter name, age and gender for every seat	tickets are issued correctly for each traveller
US-16	passenger	to pay online using UPI, card or net banking	I can confirm my booking without visiting a counter
US-17	passenger	to retry a failed payment within the lock time	I do not lose my seats because of a temporary error
US-18	passenger	to receive an e-ticket with a unique QR code	I can board without a paper ticket
US-19	passenger	to view my past and upcoming bookings	I can manage my travel plans in one place

E5 · Cancellation, Refund & Waitlist			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-20	passenger	to cancel a booked ticket online	I do not have to visit the counter when plans change
US-21	passenger	the refund to be calculated automatically from time-based slabs	I know exactly how much I will get back
US-22	passenger	to join a waitlist for a full bus and be confirmed automatically if a seat frees up	I still get a chance to travel on my preferred bus

E6 · Notifications & Tracking			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-23	passenger	SMS and email confirmations and reminders	I never miss my travel details
US-24	passenger	to track my bus live on a map with an estimated arrival time	I can reach my boarding point on time
US-25	passenger	to receive delay alerts	I can adjust my plans early

E7 · Operations & Reporting			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-26	conductor	to scan a passenger's QR code	I can verify tickets instantly and prevent fake tickets
US-27	conductor	to update the trip status (started, delayed, completed)	passengers and operators always see the latest status
US-28	bus operator (admin)	a dashboard showing sales, occupancy and cancellations	I can make informed decisions on fares and schedules
US-29	admin	to export reports as PDF or Excel	I can share them with management and authorities

E8 · Security, Quality & Operations			
ID	**As a <user>**	**I want <feature>**	**So that <benefit>**
US-30	system owner	audit logs and security hardening (encryption, input validation)	user data and payments remain protected
US-31	system owner	load and performance testing before launch	the system stays fast during festival rushes
US-32	support engineer	monitoring and alerts for the live system	outages are detected and fixed quickly

3. Define Acceptance Criteria

Acceptance criteria tell the team, and the Product Owner, exactly when a story is finished. They were written for the 8 user stories selected for Sprint 1 (see Section 5), in the **Given–When–Then** format. Each criterion is specific, measurable and testable, and each is tied to its story ID.

How to read Given–When–Then

Given describes the starting situation. **When** describes the action taken. **Then** describes the result that must be seen and measured.

US-01 · Register as passenger (5 pts)			
Scenario	**Given**	**When**	**Then**
Successful registration	I am a new visitor on the registration page	I enter a valid name, email, 10-digit mobile number and a password of at least 8 characters with a number and a symbol, and submit	my account is created in under 3 seconds and an OTP reaches my mobile within 30 seconds
OTP verification	I have received the 6-digit OTP	I enter the correct OTP within 5 minutes	my account status becomes Verified and I am taken to my home page
Duplicate account	an account already exists with my email or mobile number	I try to register again with the same email or mobile number	the system rejects the request, shows 'Already registered' and creates no second account

US-02 · Login with password or OTP (5 pts)			
Scenario	**Given**	**When**	**Then**
Valid login	I am a verified user on the login page	I enter my correct email and password	I am logged in within 2 seconds and see my home page
Account lockout	I am on the login page	I enter a wrong password 5 times in a row	my account is locked for 15 minutes and the screen shows the time remaining
OTP login	I choose 'Login with OTP'	I enter the valid OTP within 5 minutes (or an expired OTP after 5 minutes)	I am logged in with a valid OTP, and an expired OTP is rejected with a clear message

US-03 · Reset password via OTP (3 pts)			
Scenario	**Given**	**When**	**Then**
Request reset	I have forgotten my password	I submit my registered email or mobile number	a reset OTP valid for 10 minutes is sent within 30 seconds
Set new password	I hold a valid reset OTP	I enter a new password that follows the password rules	the password is updated, all old sessions are logged out and I can log in with the new password

US-04 · Manage profile (2 pts)			
Scenario	**Given**	**When**	**Then**
Update details	I am logged in and on my profile page	I change my name or mobile number and save	the change is saved within 2 seconds, a confirmation appears, and a new mobile number is accepted only after OTP verification
Invalid input	I am editing my profile	I enter a mobile number that is not 10 digits	an error message is shown and my saved details stay unchanged

US-05 · Role-based admin access (3 pts)			
Scenario	**Given**	**When**	**Then**
Passenger blocked	I am logged in as a passenger	I open an admin page address	access is denied with a 403 message and the attempt is recorded
Admin allowed	I am logged in as an admin	I open the admin panel	I see the menu for buses, routes and trips
Session timeout	my admin session has been idle for 15 minutes	I click any admin action	I am asked to log in again

US-06 · Manage buses and seat layout (5 pts)			
Scenario	**Given**	**When**	**Then**
Add bus	I am an admin on the Add Bus page	I enter a registration number, bus type and a 2x2 layout of 40 seats and save	the bus is saved and the preview shows 40 correctly numbered seats
Duplicate bus	a bus with the same registration number exists	I try to add it again	the system rejects it and shows 'Bus already exists'
Protected delete	a bus has scheduled trips	I try to delete it	the system blocks deletion and offers to mark the bus Inactive instead

US-07 · Create routes and boarding points (3 pts)			
Scenario	**Given**	**When**	**Then**
Create route	I am an admin on the Create Route page	I enter source, destination, distance and at least one boarding point with its time offset and save	the route is saved and appears in the route list
Invalid route	I am creating a route	I choose the same city as source and destination	the system rejects it with a clear error message

US-08 · Schedule trips and fares (8 pts)			
Scenario	**Given**	**When**	**Then**
Schedule trip	an active bus and a route exist	I schedule a trip with date, departure and arrival time and a base fare	the trip is created with status Scheduled and appears in the trip list
Bus overlap	a bus is already assigned from 08:00 to 14:00	I schedule the same bus at 12:00 on the same day	the system rejects it with the message 'Bus already assigned'
Recurring trips	I choose 'Daily for 30 days'	I save the schedule	30 trips with the same fare are created in under 5 seconds
Fare check	I am entering a fare	I enter a fare of zero or above 10,000	the system shows a validation error and does not save the trip

Summary: 22 acceptance scenarios cover all 8 Sprint 1 stories. The QA team turns each scenario into a test case, and a story is accepted only when all its scenarios pass.

4. Prioritize the Product Backlog

4.1 Prioritization Method

The product backlog was ordered using four factors. Together they balance what is valuable for the business and customers with what must technically come first.

Factor	Question asked	How it was used
Business value	How much does this story increase revenue, trust or efficiency?	Rated High / Medium / Low with the Product Owner.
Customer need (MoSCoW)	Can the product be used without it?	Must have, Should have or Could have. Won't have (this release): international routes, multi-currency and hotel / cab booking.
Dependencies	Which stories must be finished first?	A story that unblocks others is scheduled earlier, even if its own value is lower.
Risk and effort	How uncertain or large is it?	Risky stories (seat lock, payment) start early so problems show up early.

Ordering rule

Rank 1 is built first. Stories are ordered by (1) dependencies, (2) Must before Should before Could, (3) higher business value first, and (4) higher risk earlier.

4.2 Prioritized Product Backlog

Rank	ID	Story	Epic	Value	MoSCoW	Depends on	Pts	Sprint
1	US-01	Register as passenger	E1	High	Must	-	5	Sprint 1
2	US-02	Login with password or OTP	E1	High	Must	US-01	5	Sprint 1
3	US-03	Reset password via OTP	E1	Medium	Should	US-01	3	Sprint 1
4	US-04	Manage profile	E1	Medium	Should	US-02	2	Sprint 1
5	US-05	Role-based admin access	E2	High	Must	US-02	3	Sprint 1
6	US-06	Manage buses and seat layout	E2	High	Must	US-05	5	Sprint 1
7	US-07	Create routes and boarding points	E2	High	Must	US-05	3	Sprint 1
8	US-08	Schedule trips and fares	E2	High	Must	US-06, US-07	8	Sprint 1
9	US-09	Search buses	E3	High	Must	US-08	5	Sprint 2

Rank	ID	Story	Epic	Value	MoSCoW	Depends on	Pts	Sprint
10	US-10	Filter and sort results	E3	Medium	Should	US-09	3	Sprint 2
11	US-11	Live seat map	E3	High	Must	US-09	5	Sprint 2
12	US-12	Select and lock seats (8 min)	E3	High	Must	US-11	8	Sprint 2
13	US-13	Women-safe seat rule	E3	Medium	Should	US-11	5	Sprint 2
14	US-14	Choose boarding point	E3	Medium	Should	US-07, US-12	3	Sprint 2
15	US-15	Enter traveller details	E4	High	Must	US-12	3	Sprint 3
16	US-16	Pay online	E4	High	Must	US-15	8	Sprint 3
17	US-17	Retry failed payment	E4	High	Should	US-16	5	Sprint 3
18	US-18	E-ticket with QR code	E4	High	Must	US-16	5	Sprint 3
19	US-19	Booking history	E4	Medium	Should	US-18	3	Sprint 3
20	US-20	Cancel ticket	E5	High	Must	US-18	5	Sprint 4
21	US-21	Slab-based automatic refund	E5	High	Must	US-20	5	Sprint 4
22	US-22	Waitlist with auto-promotion	E5	Medium	Could	US-20	8	Sprint 4
23	US-23	SMS / email alerts	E6	Medium	Should	US-18	3	Sprint 4
24	US-24	Live bus tracking	E6	Medium	Could	US-08	8	Sprint 5
25	US-25	Delay alerts	E6	Low	Could	US-24	3	Sprint 5
26	US-26	Conductor QR scan	E7	High	Must	US-18	5	Sprint 5
27	US-27	Trip status update	E7	Medium	Should	US-26	3	Sprint 5
28	US-28	Admin dashboard	E7	Medium	Should	US-18	5	Sprint 5
29	US-29	Export reports	E7	Low	Could	US-28	2	Sprint 5

Rank	ID	Story	Epic	Value	MoSCoW	Depends on	Pts	Sprint
30	US-30	Audit logs and security hardening	E8	High	Must	US-02	5	Sprint 6
31	US-31	Load and performance testing	E8	Medium	Should	US-16	5	Sprint 6
32	US-32	Monitoring and alerts	E8	Medium	Should	US-30	5	Sprint 6

Shading alternates with each sprint to show how the backlog is grouped. Sprints 2 to 6 are planned provisionally and will be refined after each Sprint Review.

4.3 Work Load by Sprint and Priority

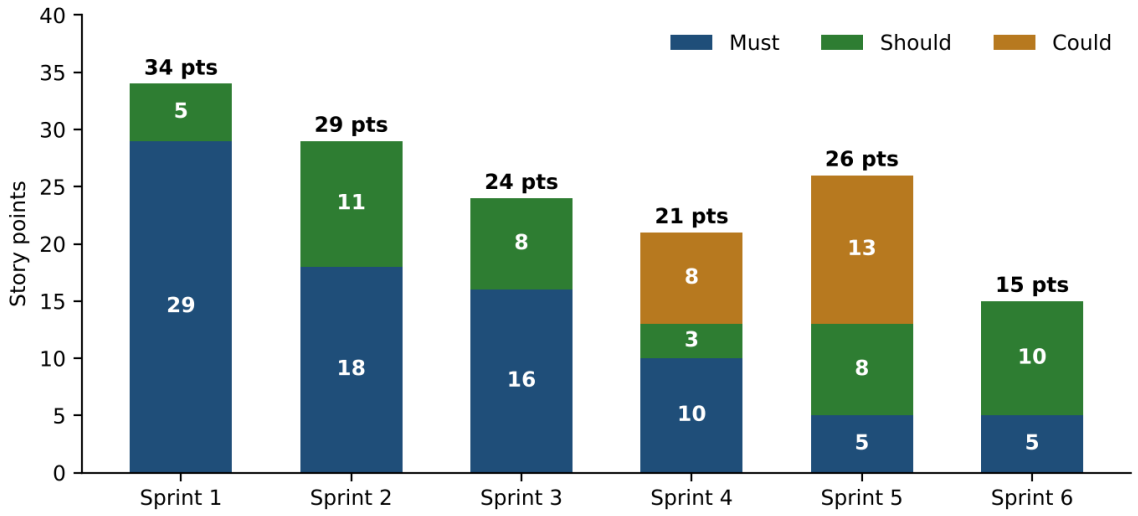


Figure 3: *Story points per sprint, split by MoSCoW priority*

Almost all Must-have stories fall in Sprints 1 to 3, so the core booking journey is usable early. Could-have stories (waitlist, live tracking, delay alerts and report export) sit in Sprints 4 and 5 and can be dropped without breaking the product if time runs short.

Why Sprint 1 starts with accounts and fleet set-up

The most valuable and riskiest feature is the 8-minute seat lock (US-12), but it cannot be built until buses, routes and trips exist (US-06 to US-08) and users can log in (US-01, US-02). Sprint 1 therefore builds the foundation that every later story depends on, and delivers a working admin set-up that the operator can use immediately.

5. Plan the Sprint Backlog

5.1 Sprint 1 at a Glance

Item	Detail
Sprint	Sprint 1 of 6 (foundation sprint)
Length	2 weeks (10 working days)
Team	5 developers (2 backend, 2 frontend, 1 QA), a Product Owner and a Scrum Master
Capacity	300 productive hours (5 people × 10 days × 6 focused hours per day)
Commitment	8 user stories, 34 story points, 287 task hours
Sprint goal	Users can register and log in securely, and the admin can set up buses, routes and scheduled trips with fares (see Section 7).

5.2 Selected User Stories

The eight highest-ranked stories in the backlog were pulled into Sprint 1. They fit within the team's capacity and form one complete, demonstrable slice: sign-up and login for users, and set-up of buses, routes and trips for admins.

ID	Story	Pts	Reason for selecting it now
US-01	Register as passenger	5	Starting point for every user journey.
US-02	Login with password or OTP	5	Needed by every feature that requires a user.
US-03	Reset password via OTP	3	Completes the account feature; reuses OTP work.
US-04	Manage profile	2	Small story that completes account management.
US-05	Role-based admin access	3	Protects all admin screens; needed before set-up stories.
US-06	Manage buses and seat layout	5	Buses and seat layouts are needed for the seat map in Sprint 2.
US-07	Create routes and boarding points	3	Routes and boarding points are needed for search in Sprint 2.
US-08	Schedule trips and fares	8	Trips with fares are the data that Sprint 2 search will show.
	Total	**34**	

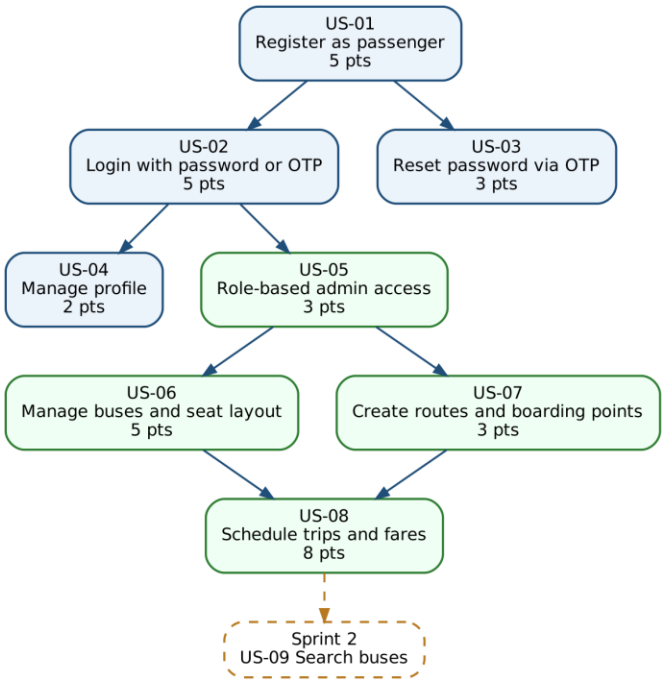


Figure 4: *Dependencies between Sprint 1 stories*

Reading the dependency map

An arrow means "must be finished before". It decides the order of work inside the sprint.

- **US-01** is the root: nothing else can start without user accounts.
- **US-02** unlocks profile (US-04) and admin access (US-05).
- **US-05** unlocks bus and route set-up (US-06, US-07), which can run in parallel.
- **US-08** needs both buses and routes, so it starts last but is the largest story.
- The dashed box shows the Sprint 2 story that depends on US-08.

5.3 Task Breakdown

Each story was divided into small development tasks of roughly 4 to 14 hours. Common set-up and team-wide tasks are listed separately so that the capacity check is honest.

Task	Story	Task description	Owner	Hours
T0.1	Enabler	Set up repository, branching rules and CI pipeline	DevOps	8
T0.2	Enabler	Create database schema for users, buses, routes and trips	Backend	10
T0.3	Enabler	Build base UI layout and navigation	Frontend	8
T01.1	US-01	Design registration form with validation messages	Frontend	6
T01.2	US-01	Registration API with validation and password hashing	Backend	10
T01.3	US-01	Mobile / email OTP verification	Backend	8
T01.4	US-01	Unit and API tests for registration	QA	6
T02.1	US-02	Login screen with password and OTP options	Frontend	5
T02.2	US-02	Authentication API with token session and 5-attempt lockout	Backend	10
T02.3	US-02	OTP login flow	Backend	6
T02.4	US-02	Tests for login, lockout and OTP expiry	QA	5
T03.1	US-03	Forgot-password screens	Frontend	4
T03.2	US-03	Reset OTP and new-password API	Backend	7

Task	Story	Task description	Owner	Hours
T03.3	US-03	Tests for reset flow	QA	3
T04.1	US-04	Profile page	Frontend	4
T04.2	US-04	Profile update API with phone re-verification	Backend	4
T04.3	US-04	Tests for profile update	QA	2
T05.1	US-05	Role and permission model	Backend	6
T05.2	US-05	Route guards for admin pages	Frontend	5
T05.3	US-05	Tests for access control	QA	4
T06.1	US-06	Add / edit bus form with seat-layout preview	Frontend	8
T06.2	US-06	Bus create-read-update-deactivate API	Backend	10
T06.3	US-06	Seat-layout generator (2x2 seater and sleeper)	Backend	8
T06.4	US-06	Tests for bus management	QA	6
T07.1	US-07	Route form with ordered boarding points	Frontend	6
T07.2	US-07	Route create-read-update API	Backend	8
T07.3	US-07	Tests for route management	QA	4
T08.1	US-08	Trip scheduling screen with calendar and recurrence	Frontend	12
T08.2	US-08	Trip API with bus-overlap check	Backend	14
T08.3	US-08	Fare setting by seat type	Backend	8
T08.4	US-08	Link trips with bus and route data	Backend	6
T08.5	US-08	Tests for scheduling, overlap and recurrence	QA	10
TS.1	Team	Code reviews and pair programming	Team	20
TS.2	Team	Test data, staging environment and deployment	DevOps	10
TS.3	Team	API documentation	Backend	10
TS.4	Team	Bug-fix buffer	Team	20
TS.5	Team	Sprint review demo preparation	Team	6
		Total task hours		**287**

Hours by role:

Backend	Frontend	QA	DevOps	Team
125 h	58 h	40 h	18 h	46 h

The 287 planned hours use 96% of the 300-hour capacity and leave 13 hours as a safety margin.

5.4 Sprint 1 Schedule

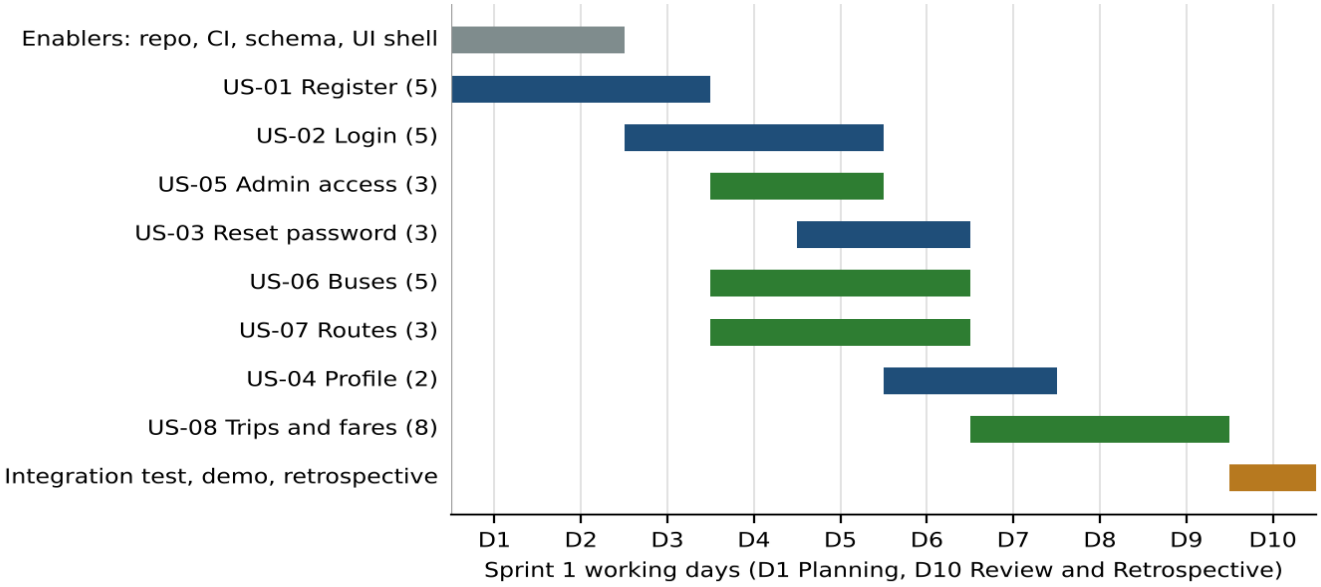


Figure 5: *Planned order of work across the 10 sprint days*

5.5 Definition of Done

A story is counted as finished only when every point below is true:

- Code is written, peer-reviewed and merged to the main branch.
- Unit and API tests pass, with at least 80% coverage on new code.
- Every acceptance criterion of the story has been verified by QA.
- No open critical or high-severity bugs remain.
- The feature is deployed to the staging environment and API documentation is updated.
- The Product Owner has seen the feature in the Sprint Review and accepted it.

6. Estimate Effort Using Story Points

6.1 Estimation Technique

Story points measure the **relative size** of a story, combining effort, complexity and uncertainty, instead of counting hours. The team used **Planning Poker** with the **Fibonacci scale** (1, 2, 3, 5, 8, 13). The gaps between numbers grow on purpose, because larger stories are harder to estimate precisely.

- **Baseline:** US-04 (Manage profile) was agreed as the reference story and given 2 points. All other stories were sized against it.
- **Process:** the Product Owner explains a story; every member picks a card in secret; cards are revealed together. The highest and lowest voters explain their reasons and the team votes again until the estimates agree.
- **Rule:** any story estimated above 8 points is split into smaller stories before it can enter a sprint.

6.2 Planning Poker Results

Story	Title	Vote 1	Vote 2	Vote 3	Vote 4	Vote 5	Agreed
US-01	Register as passenger	5	5	3	8	5	**5**
US-02	Login with password or OTP	5	5	5	3	8	**5**
US-03	Reset password via OTP	3	2	3	3	5	**3**
US-04	Manage profile	2	2	1	2	3	**2**
US-05	Role-based admin access	3	3	5	3	3	**3**
US-06	Manage buses and seat layout	5	8	5	5	3	**5**
US-07	Create routes and boarding points	3	3	2	3	5	**3**
US-08	Schedule trips and fares	8	13	8	8	5	**8**

Votes show the first round only. The agreed value is the result after discussion, as described above.

6.3 Justification of Estimates

Story	Title	Pts	Why this estimate
US-01	Register as passenger	5	Form, validation, password hashing and OTP integration. Moderate complexity; the OTP provider is new to the team.
US-02	Login with password or OTP	5	Session handling, lockout rule and OTP login. Security-sensitive, so it needs careful testing.
US-03	Reset password via OTP	3	Reuses the OTP and password logic from US-01; one new API and two small screens.
US-04	Manage profile	2	Simple profile page and update API. Chosen as the baseline (reference) story worth 2 points.
US-05	Role-based admin access	3	Small role model and route guards, but it touches every admin page.
US-06	Manage buses and seat layout	5	CRUD plus a seat-layout generator. The layout logic is the uncertain part.
US-07	Create routes and boarding points	3	CRUD with ordered boarding points. Well understood work.
US-08	Schedule trips and fares	8	Most complex story: overlap check, recurring trips and fare rules. Depends on US-06 and US-07. Split further if it grows beyond 8.

6.4 Capacity Check and Burndown

Measure	Value
Points committed	34 story points
Assumed velocity	About 34 points (no history yet; estimated at about 0.7 points per person-day for 5 people over 10 days)
Task hours committed	287 of 300 available hours
Safety margin	13 hours
Stories to drop if needed	US-04 (2 pts) first, then US-03 (3 pts); both are Should-have

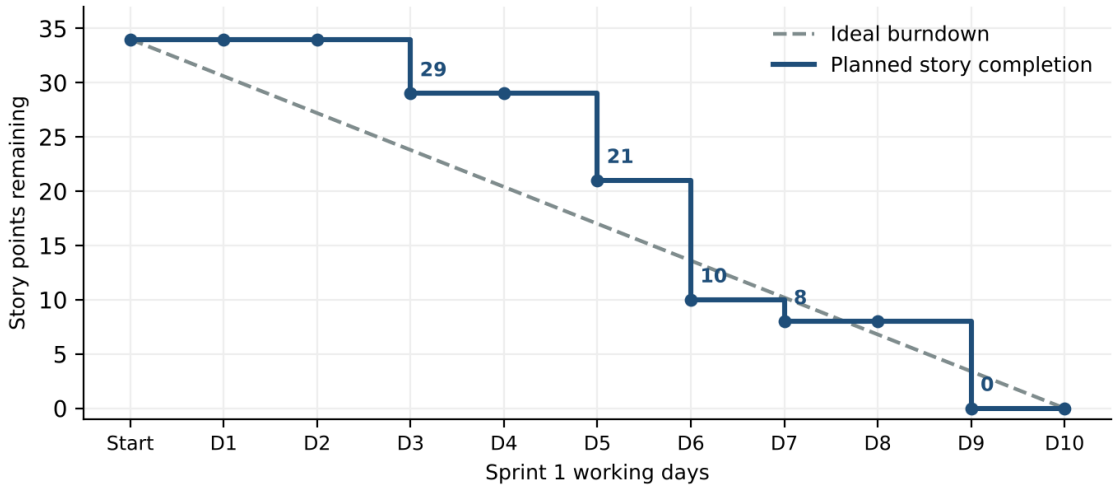


Figure 6: *Planned Sprint 1 burndown against the ideal line*

The step line shows when each story is planned to be completed. The team will update the real burndown every day at the Daily Scrum and compare it with this plan.

7. Present the Sprint Goal and Expected Deliverables

7.1 Sprint Goal

Sprint 1 Goal

By the end of Sprint 1, passengers can securely register, verify and log in to BusBuddy, and the admin can set up buses, routes and scheduled trips with fares, so that the search and seat-selection features of Sprint 2 will have real data to work with.

7.2 Expected Deliverables

#	Deliverable	What it includes	How it is verified
1	Account module	Registration with OTP, login (password and OTP), password reset and profile page	All acceptance scenarios of US-01 to US-04 pass
2	Role-based admin panel	Admin-only access, 15-minute session timeout	Access tests for passenger and admin roles
3	Bus and seat-layout management	Add, edit and deactivate buses with a 2×2 or sleeper layout preview	Demo and QA test cases (US-06)
4	Route management	Routes with source, destination and boarding points	Demo and QA test cases (US-07)
5	Trip scheduling with fares	Single and daily recurring trips, overlap check, fare validation	Demo and QA test cases (US-08)
6	Technical foundation	Database schema v1, CI pipeline and a staging environment	Pipeline runs green; staging link available
7	Quality evidence	Test report, 80% or higher coverage on new code, API documentation	Test report reviewed by Scrum Master
8	Sprint records	Updated backlog, review notes and retrospective actions	Shared with all stakeholders

7.3 Features Completed at the End of the Sprint

- A new user can create an account, verify the mobile number with an OTP, log in, reset a forgotten password and edit the profile.
- An admin can log in, add buses with seat layouts, create routes with boarding points, and schedule trips with fares, including daily recurring trips.
- Passengers cannot open admin pages, and admin sessions expire after 15 minutes of inactivity.
- All of the above is deployed on the staging environment and ready to demonstrate.

7.4 Sprint Events

Event	When	Time-box	Purpose
Sprint Planning	Day 1	2 hours	Agree the goal, select stories and break them into tasks.
Daily Scrum	Every day	15 minutes	Share progress and obstacles; update the burndown.
Backlog Refinement	Day 6	1 hour	Clarify and estimate stories for Sprint 2.
Sprint Review	Day 10	1 hour	Demonstrate the work and collect feedback from stakeholders.
Sprint Retrospective	Day 10	45 minutes	Decide what to improve in the next sprint.

7.5 Sprint Risks and Responses

Risk	Impact	Response
OTP / SMS provider set-up takes longer than planned	High	Use the provider's sandbox first and a mock OTP service for testing.
US-08 (8 points) grows larger than estimated	Medium	Start it as soon as US-06 and US-07 finish; split recurring trips into a separate story if needed.
Team is new to the chosen tools	Medium	Pair programming and code reviews in the first week.
Requirements change during the sprint	Low	New ideas go to the product backlog and are considered for Sprint 2.

Summary

This workshop turned the BusBuddy vision into 8 epics and 32 user stories, ordered them by value, dependencies and risk, and planned Sprint 1 with 8 stories (34 points, 287 hours). The sprint goal is clear, the deliverables are demonstrable, and the plan stays inside the team's capacity.